

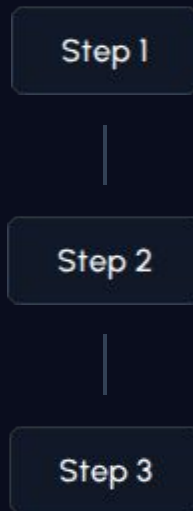
Visual Logic

The Fundamentals of Conditionals

PROGRAM FLOW CONTROL

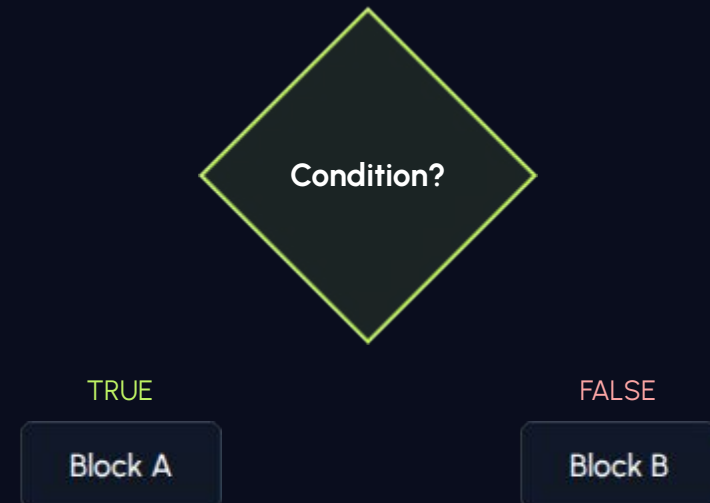
Fixed Flow

Executes the same steps every time.



Branching Flow

Logic decides path based on state.



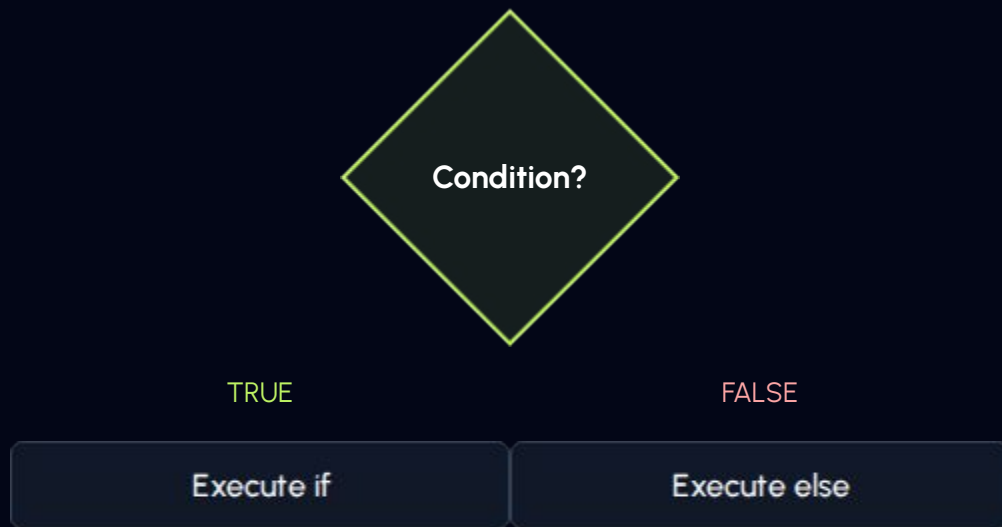
| THE "IF" STATEMENT



Only the **indented block** is conditional. Code after the indentation always executes.

```
if temperature < 20:  
    print("Cold")  
  
print("Always")
```

THE "IF-ELSE" STATEMENT



A binary choice: You always take exactly one path.



| MULTIPLE PATH CHAINS



Execution **short-circuits**: as soon as one condition is met, the rest of the chain is skipped.

| COMPARISON TOOLS

Operator	Meaning	Example
==	Equal to	x == 5
!=	Not equal to	x != 10
> / <	Greater / Less Than	age > 18
>= / <=	Greater/Less or Equal	score >= 70

| COMBINING CONDITIONS



AND

Both must be
True.



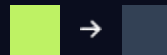
OR

At least one is
True.

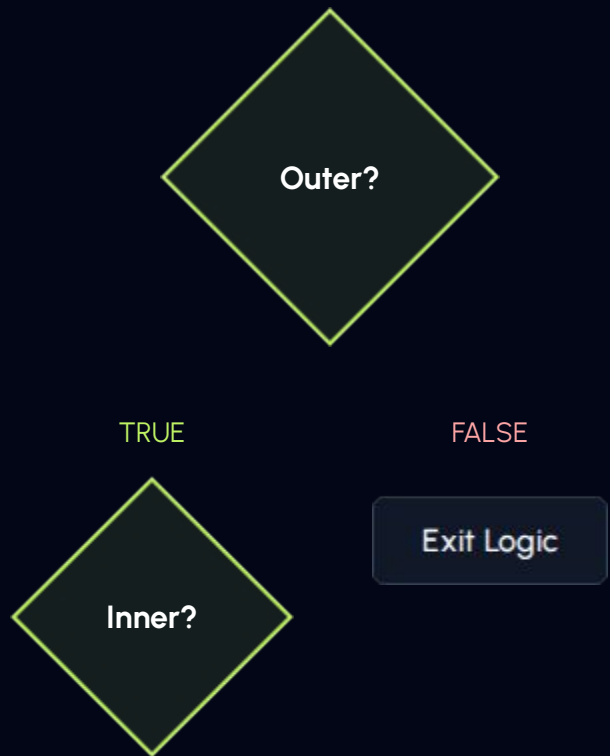


NOT

Flips the truth
value.



LAYERED LOGIC



The Club Analogy

1. age $\geq$ 18? (Outer)
2. has_id? (Inner)

"The inner decision only makes sense if the outer decision is true."

| GUARD CLAUSE PATTERN

Bad: Nested

Creates "Arrow Code" that is hard to read.

```
if cond:
    if other:
        if another:
            do_work()
```

Good: Guard

Early returns keep the main logic flat.

```
if not cond: return
if not other: return
if not another: return

do_work()
```

Ternary Expression

```
value_if_true if condition else value_if_false
```

```
status = "adult" if age >= 18 else "minor"
```

| COMMON MISTAKES

Assignment

Using = instead of == in comparisons.

The Colon

Forgetting the : after the condition header.

Indentation

Inconsistent spacing leads to scope errors.

TRACE VISUALIZATION

Input: 5

```
1. number > 0? True
   → print("Positive")
   → 5 % 2 == 0? False
      → print("and odd")
```

Input: -3

```
1. number > 0? False
2. number < 0? True
   → print("Negative")
```

| IMAGE SOURCES



<https://static.vecteezy.com/system/resources/thumbnails/079/062/385/large/abstract-binary-code-pattern-on-dark-background-free-video.jpg>

Source: www.vecteezy.com